

Ejercicio Clase 9

Consigna: Por cada ejercicio, escribir el código y agregar una captura de pantalla del resultado obtenido.

1. Crear una base de datos en Hive llamada northwind_analytics

```
hive> create database northwind_analytics
> ;
OK
Time taken: 2.133 seconds
hive> show databases;
OK
default
f1
northwind_analytics
tripdata
trips
Time taken: 0.058 seconds, Fetched: 5 row(s)
hive>
```

2. Crear un script para importar un archivo .parquet de la base northwind que contenga la lista de clientes junto a la cantidad de productos vendidos ordenados de mayor a menor (campos customer_id, company_name, productos_vendidos). Luego ingestar el archivo a HDFS (carpeta /sqoop/ingest/clientes). Pasar la password en un archivo

Primero paso la password de acceso a la base **northwind** a un archivo de texto plano oculto: **.sqoop_pw** luego cambio los permisos.

```
hadoop@615cf53bef6c:/$ # 1) Guardo la password en un archivo (no en la línea de comandos):
hadoop@615cf53bef6c:/$
hadoop@615cf53bef6c:/$ echo -n 'edvai' > ~/.sqoop_pw
hadoop@615cf53bef6c:/$ chmod 600 ~/.sqoop_pw
```

Me aseguro que el Driver JDBC esta en la carpeta de Sqoop:

```
hadoop@615cf53bef6c:~$ ls /usr/lib/sqoop/lib/postgresql-*.jar
me/hadoop/jars/postgresql-42.7.4.jar
/usr/lib/sqoop/lib/postgresql-42.3.3.jar
```

Creo el archivo **query_db_2_hdfs_parquet.sh** para hacer la importación con sqoop:

```

hadoop@615cf53bef6c:~/scripts$ cat > query_db_2_hdfs_parquet.sh
sqoop import \
  --connect "jdbc:postgresql://172.17.0.3:5432/northwind" \
  --username postgres \
  --password-file file:///home/hadoop/.sqoop_pw \
  --driver org.postgresql.Driver \
  --query "
    SELECT
      c.customer_id,
      c.company_name,
      COALESCE(SUM(od.quantity), 0) AS productos_vendidos
    FROM customers c
    LEFT JOIN orders o
      ON o.customer_id = c.customer_id
    LEFT JOIN order_details od
      ON od.order_id = o.order_id
    WHERE \$CONDITIONS
    GROUP BY c.customer_id, c.company_name
    ORDER BY productos_vendidos DESC
  " \
  --as-parquetfile \
  --target-dir /sqoop/ingest/clientes \
  --delete-target-dir \
  --num-mappers 1
hadoop@615cf53bef6c:~/scripts$

```

Cambio permisos para que pueda ser ejecutado:

```

hadoop@615cf53bef6c:~/scripts$ chmod 777 query_db_2_hdfs_parquet.sh
hadoop@615cf53bef6c:~/scripts$ ls -l
total 64
drwxrwxr-x 3 hadoop hadoop 4096 Sep 29 19:46 Notebook
-rw-rw-r-- 1 hadoop hadoop 670 Feb 28 2022 derby.log
-rwxrwxr-x 1 hadoop hadoop 272 Sep 22 01:14 ingest.sh
-r-xr-xr-x 1 hadoop hadoop 398 Sep 17 16:57 landing.sh
-rwxrwxrwx 1 hadoop hadoop 716 Oct 18 16:24 landing_2.sh
-rwxrwxrwx 1 hadoop hadoop 914 Oct 24 19:25 landing_3.sh
-rwxrwxrwx 1 hadoop hadoop 273 Sep 28 12:14 pyspark_jupyter.sh
-rwxrwxrwx 1 hadoop hadoop 668 Nov 1 15:12 query_db_2_hdfs_parquet.sh
drwxr-xr-x 2 hadoop hadoop 4096 Feb 9 2022 spark-warehouse
-rwxrwxrwx 1 hadoop hadoop 1089 May 9 2022 start-services.sh
-rwxrwxrwx 1 hadoop hadoop 1964 Oct 19 00:37 transform_load.py
-rwxrwxrwx 1 hadoop hadoop 2897 Oct 19 12:48 transform_load_2.py
-rwxrwxrwx 1 hadoop hadoop 3909 Oct 25 04:35 transform_load_3.py
-rwxrwxrwx 1 hadoop hadoop 1058 May 9 2022 transformation.py
-rwxrwxrwx 1 hadoop hadoop 2068 Oct 31 22:14 weather_str_transform.py
-rwxrwxrwx 1 hadoop hadoop 1747 Nov 1 01:13 yahoo_weather2kafka.py
hadoop@615cf53bef6c:~/scripts$

```

Pruebo el .sh para ver si funciona correctamente:

```

hadoop@615cf53bef6c:~/scripts$ ./query_db_2_hdfs_parquet.sh
Warning: /usr/lib/sqoop/./hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /usr/lib/sqoop/./hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /usr/lib/sqoop/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /usr/lib/sqoop/./zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2025-11-01 15:16:12,107 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2025-11-01 15:16:12,979 WARN sqoop.ConnFactory: Parameter --driver is set to an explicit driver however appropriate connection manager is not being set (via --connection-manager). Sqoop is going to fall back to org.apache.sqoop.manager.GenericJdbcManager. Please specify explicitly which connection manager should be used next time.
2025-11-01 15:16:12,998 INFO manager.SqlManager: Using default fetchSize of 1000
2025-11-01 15:16:12,998 INFO tool.CodeGenTool: Beginning code generation

```

(Faltan algunas pantallas. Visualizo acá solo el inicio y el final)

```

Map-Reduce Framework
  Map input records=91
  Map output records=91
  Input split bytes=87
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=71
  CPU time spent (ms)=5480
  Physical memory (bytes) snapshot=254394368
  Virtual memory (bytes) snapshot=2998194176
  Total committed heap usage (bytes)=161480704
  Peak Map Physical memory (bytes)=254394368
  Peak Map Virtual memory (bytes)=2998411264
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=0
2025-11-01 15:16:55,888 INFO mapreduce.ImportJobBase: Transferred 5.2852 KB in 34.795 seconds (155.5397 bytes/sec)
2025-11-01 15:16:55,897 INFO mapreduce.ImportJobBase: Retrieved 91 records.
hadoop@615cf53bef6c:~/scripts$

```

El script funciona correctamente. Verifico que el archivo parquet este alojado donde corresponde en hdfs:

```

hadoop@615cf53bef6c:~/scripts$ hdfs dfs -ls /sqoop/ingest/clientes
Found 3 items
drwxr-xr-x   - hadoop supergroup          0 2025-11-01 15:16 /sqoop/ingest/clientes/.metadata
drwxr-xr-x   - hadoop supergroup          0 2025-11-01 15:16 /sqoop/ingest/clientes/.signals
-rw-r--r--   1 hadoop supergroup    3916 2025-11-01 15:16 /sqoop/ingest/clientes/1ad46101-8db7-4422-8b51-2a3e0c1c9842.parquet
hadoop@615cf53bef6c:~/scripts$

```

3. Crear un script para importar un archivo .parquet de la base northwind que contenga la lista de órdenes junto a qué empresa realizó cada pedido (campos order_id, shipped_date, company_name, phone). Luego ingestar el archivo a HDFS (carpeta /sqoop/ingest/envíos). Pasar la password en un archivo

Realizando el mismo procedimiento que recién, pero cambiando la query, creo el archivo **query_db_2_hdfs_parquet_2.sh** para hacer la importación con sqoop (aunque la consigna no lo pide, ordeno por customer_id y luego dentro de eso, por shipped_date):

```

hadoop@615cf53bef6c:~/scripts$ cat > query_db_2_hdfs_parquet_2.sh
sqoop import \
  --connect "jdbc:postgresql://172.17.0.3:5432/northwind" \
  --username postgres \
  --password-file file:///home/hadoop/.sqoop_pw \
  --driver org.postgresql.Driver \
  --query "
    select
      o.order_id,
      o.shipped_date,
      c.company_name,
      c.phone
    from orders o
    left join customers c
      on o.customer_id = c.customer_id
    WHERE \$CONDITIONS
    order by c.customer_id, o.shipped_date asc
  " \
  --as-parquetfile \
  --target-dir /sqoop/ingest/envios \
  --delete-target-dir \
  --num-mappers 1
hadoop@615cf53bef6c:~/scripts$

```

Cambio permisos:

```

hadoop@615cf53bef6c:~/scripts$ chmod 777 query_db_2_hdfs_parquet_2.sh
hadoop@615cf53bef6c:~/scripts$ ls -l
total 84
drwxrwxr-x 3 hadoop hadoop 4096 Sep 29 19:46 Notebook
-rw-rw-r-- 1 hadoop hadoop 14298 Nov  1 15:16 QueryResult.java
-rw-rw-r-- 1 hadoop hadoop 670 Feb 28 2022 derby.log
-rwxrwxr-x 1 hadoop hadoop 272 Sep 22 01:14 ingest.sh
-r-xr-xr-x 1 hadoop hadoop 398 Sep 17 16:57 landing.sh
-rwxrwxrwx 1 hadoop hadoop 716 Oct 18 16:24 landing_2.sh
-rwxrwxrwx 1 hadoop hadoop 914 Oct 24 19:25 landing_3.sh
-rwxrwxrwx 1 hadoop hadoop 273 Sep 28 12:14 pyspark_jupyter.sh
-rwxrwxrwx 1 hadoop hadoop 668 Nov  1 15:12 query_db_2_hdfs_parquet.sh
-rwxrwxrwx 1 hadoop hadoop 549 Nov  1 16:00 query_db_2_hdfs_parquet_2.sh
drwxr-xr-x 2 hadoop hadoop 4096 Feb  9 2022 spark-warehouse
-rwxrwxrwx 1 hadoop hadoop 1089 May  9 2022 start-services.sh
-rwxrwxrwx 1 hadoop hadoop 1964 Oct 19 00:37 transform_load.py
-rwxrwxrwx 1 hadoop hadoop 2897 Oct 19 12:48 transform_load_2.py
-rwxrwxrwx 1 hadoop hadoop 3909 Oct 25 04:35 transform_load_3.py
-rwxrwxrwx 1 hadoop hadoop 1058 May  9 2022 transformation.py
-rwxrwxrwx 1 hadoop hadoop 2068 Oct 31 22:14 weather_str_transform.py
-rwxrwxrwx 1 hadoop hadoop 1747 Nov  1 01:13 yahoo_weather2kafka.py
hadoop@615cf53bef6c:~/scripts$

```

Verifico que el script genero el archivo parquet en destino:

```
hadoop@615cf53bef6c:~/scripts$ hdfs dfs -ls /sqoop/ingest/envios
Found 3 items
drwxr-xr-x - hadoop supergroup          0 2025-11-01 15:55 /sqoop/ingest/envios/.metadata
drwxr-xr-x - hadoop supergroup          0 2025-11-01 15:56 /sqoop/ingest/envios/.signals
-rw-r--r-- 1 hadoop supergroup    11511 2025-11-01 15:56 /sqoop/ingest/envios/3f0ae773-d7e1-471d-b82e-791d8c6ff2cc.parquet
hadoop@615cf53bef6c:~/scripts$
```

EL script funciona.

4. Crear un script para importar un archivo .parquet de la base northwind que contenga la lista de detalles de órdenes (campos order_id, unit_price, quantity, discount). Luego ingestar el archivo a HDFS (carpeta /sqoop/ingest/order_details). Pasar la password en un archivo

Realizando el mismo procedimiento que recién, pero cambiando la query, creo el archivo **query_db_2_hdfs_parquet_3.sh** para hacer la importación con sqoop (aunque la consigna no lo pide, ordeno por quantity, de manera descendente):

```
hadoop@615cf53bef6c:~/scripts$ cat > query_db_2_hdfs_parquet_3.sh
sqoop import \
  --connect "jdbc:postgresql://172.17.0.3:5432/northwind" \
  --username postgres \
  --password-file file:///home/hadoop/.sqoop_pw \
  --driver org.postgresql.Driver \
  --query "
    select
      od.order_id ,
      od.unit_price ,
      od.quantity ,
      od.discount
    from order_details od
    WHERE \$CONDITIONS
    order by od.quantity desc
  " \
  --as-parquetfile \
  --target-dir /sqoop/ingest/order_details \
  --delete-target-dir \
  --num-mappers 1
hadoop@615cf53bef6c:~/scripts$ |
```

Cambio permisos:

```

hadoop@615cf53bef6c:~/scripts$ chmod 777 query_db_2_hdfs_parquet_3.sh
hadoop@615cf53bef6c:~/scripts$ ls -l
total 88
drwxrwxr-x 3 hadoop hadoop 4096 Sep 29 19:46 Notebook
-rw-rw-r-- 1 hadoop hadoop 14298 Nov 1 15:16 QueryResult.java
-rw-rw-r-- 1 hadoop hadoop 670 Feb 28 2022 derby.log
-rwxrwxr-x 1 hadoop hadoop 272 Sep 22 01:14 ingest.sh
-r-xr-xr-x 1 hadoop hadoop 398 Sep 17 16:57 landing.sh
-rwxrwxrwx 1 hadoop hadoop 716 Oct 18 16:24 landing_2.sh
-rwxrwxrwx 1 hadoop hadoop 914 Oct 24 19:25 landing_3.sh
-rwxrwxrwx 1 hadoop hadoop 273 Sep 28 12:14 pyspark_jupyter.sh
-rwxrwxrwx 1 hadoop hadoop 668 Nov 1 15:12 query_db_2_hdfs_parquet.sh
-rwxrwxrwx 1 hadoop hadoop 549 Nov 1 16:00 query_db_2_hdfs_parquet_2.sh
-rwxrwxrwx 1 hadoop hadoop 486 Nov 1 16:31 query_db_2_hdfs_parquet_3.sh
drwxr-xr-x 2 hadoop hadoop 4096 Feb 9 2022 spark-warehouse
-rwxrwxrwx 1 hadoop hadoop 1089 May 9 2022 start-services.sh
-rwxrwxrwx 1 hadoop hadoop 1964 Oct 19 00:37 transform_load.py
-rwxrwxrwx 1 hadoop hadoop 2897 Oct 19 12:48 transform_load_2.py
-rwxrwxrwx 1 hadoop hadoop 3909 Oct 25 04:35 transform_load_3.py
-rwxrwxrwx 1 hadoop hadoop 1058 May 9 2022 transformation.py
-rwxrwxrwx 1 hadoop hadoop 2068 Oct 31 22:14 weather_str_transform.py
-rwxrwxrwx 1 hadoop hadoop 1747 Nov 1 01:13 yahoo_weather2kafka.py
hadoop@615cf53bef6c:~/scripts$ |

```

Verifico que el script genero el archivo parquet en destino:

```

hadoop@615cf53bef6c:~/scripts$ hdfs dfs -ls /sqoop/ingest/order_details
Found 3 items
drwxr-xr-x - hadoop supergroup 0 2025-11-01 16:35 /sqoop/ingest/order_details/.metadata
drwxr-xr-x - hadoop supergroup 0 2025-11-01 16:35 /sqoop/ingest/order_details/.signals
-rw-r--r-- 1 hadoop supergroup 11043 2025-11-01 16:35 /sqoop/ingest/order_details/e76affc6-905e-4f16-abd7-9cd5e2e8726b.parquet
hadoop@615cf53bef6c:~/scripts$ |

```

El script funciona.

5. Generar un archivo .py que permita mediante Spark insertar en hive en la db northwind_analytics en la tabla products_sold, los datos del punto 2, pero solamente aquellas compañías en las que la cantidad de productos vendidos fue mayor al promedio.

Primero creamos la tabla **products_sold** en la base **northwind_analytics** en Hive

```

hive> CREATE EXTERNAL TABLE northwind_analytics.products_sold (customer_id string, company_name string, productos_vendidos bigint)
> STORED AS PARQUET
> LOCATION '/tables/external/northwind_analytics/products_sold';
OK
Time taken: 0.139 seconds

```

Allí insertaremos los datos de la vista generada en pyspark.

Creamos el archivo **transform_load_4.py** en **home/hadoop/scripts**

```

hadoop@615cf53bef6c:~/scripts$ cat > transform_load_4.py
#####
# Archivo: transform_load_4.py
#####

# Importamos librerias

from pyspark.sql import SparkSession

def main():
    # 1) Crear la sesión de Spark con soporte Hive

    spark = (
        SparkSession
        .builder
        .appName("transform_load_4")
        .enableHiveSupport()
        .getOrCreate()
    )

    # 2) Rutas de entrada (parquet desde HDFS)
    df = spark.read.parquet("/sqoop/ingest/clientes")

    # 3) Vistas temporales

    df.createOrReplaceTempView("clientes_cantidad_prod_vendidos_v")

    # 4) Transformaciones

    df2_load = spark.sql("SELECT * FROM clientes_cantidad_prod_vendidos_v WHERE productos_vendidos > (SELECT AVG(productos_vendidos) FROM clientes_cantidad_prod_vendidos_v)")

    # 5) Loads

    df2_load.write.mode("overwrite").parquet("/tables/external/northwind_analytics/products_sold")

    spark.stop()

if __name__ == "__main__":
    main()
hadoop@615cf53bef6c:~/scripts$

```

Cambiamos permisos:

```

hadoop@615cf53bef6c:~/scripts$ chmod 777 transform_load_4.py
hadoop@615cf53bef6c:~/scripts$ ls -l
total 92
drwxrwxr-x 3 hadoop hadoop 4096 Sep 29 19:46 Notebook
-rw-rw-r-- 1 hadoop hadoop 14298 Nov 1 15:16 QueryResult.java
-rw-rw-r-- 1 hadoop hadoop 670 Feb 28 2022 derby.log
-rwxrwxr-x 1 hadoop hadoop 272 Sep 22 01:14 ingest.sh
-r-xr-xr-x 1 hadoop hadoop 398 Sep 17 16:57 landing.sh
-rwxrwxrwx 1 hadoop hadoop 716 Oct 18 16:24 landing_2.sh
-rwxrwxrwx 1 hadoop hadoop 914 Oct 24 19:25 landing_3.sh
-rwxrwxrwx 1 hadoop hadoop 273 Sep 28 12:14 pyspark_jupyter.sh
-rwxrwxrwx 1 hadoop hadoop 668 Nov 1 15:12 query_db_2_hdfs_parquet.sh
-rwxrwxrwx 1 hadoop hadoop 549 Nov 1 16:00 query_db_2_hdfs_parquet_2.sh
-rwxrwxrwx 1 hadoop hadoop 486 Nov 1 16:31 query_db_2_hdfs_parquet_3.sh
drwxr-xr-x 2 hadoop hadoop 4096 Feb 9 2022 spark-warehouse
-rwxrwxrwx 1 hadoop hadoop 1089 May 9 2022 start-services.sh
-rwxrwxrwx 1 hadoop hadoop 1964 Oct 19 00:37 transform_load.py
-rwxrwxrwx 1 hadoop hadoop 2897 Oct 19 12:48 transform_load_2.py
-rwxrwxrwx 1 hadoop hadoop 3909 Oct 25 04:35 transform_load_3.py
-rwxrwxrwx 1 hadoop hadoop 1133 Nov 2 01:09 transform_load_4.py
-rwxrwxrwx 1 hadoop hadoop 1058 May 9 2022 transformation.py
-rwxrwxrwx 1 hadoop hadoop 2068 Oct 31 22:14 weather_str_transform.py
-rwxrwxrwx 1 hadoop hadoop 1747 Nov 1 01:13 yahoo_weather2kafka.py
hadoop@615cf53bef6c:~/scripts$

```

Luego chequeamos que el archivo funciona correctamente desde **pyspark** (antes de integrarlo al piplane)

```
hadoop@615cf53bef6c:/$ /home/hadoop/spark/bin/spark-submit --master local[*] \
> --conf spark.sql.warehouse.dir=/user/hive/warehouse \
> --conf spark.sql.catalogImplementation=hive \
> --files /home/hadoop/hive/conf/hive-site.xml \
> /home/hadoop/scripts/transform_load_4_test.py
```

Verificamos que realmente se cargaron los datos en la tabla de Hive

```
hive> select * from products_sold;
OK
SAVEA    Save-a-lot Markets      4958
ERNSH    Ernst Handel           4543
QUICK    QUICK-Stop              3961
HUNGO    Hungry Owl All-Night Grocers 1684
FRANK    Frankenversand          1525
RATTC    Rattlesnake Canyon Grocery 1383
FOLKO    Folk och fä HB              1234
HILAA    HILARION-Abastos              1096
SUPRD    Suprêmes délices             1072
WHITC    White Clover Markets          1063
QUEEN    Queen Cozinha                1031
BERGS    Berglunds snabbköp           1001
BONAP    Bon app'                      980
LINOD    LINO-Delicatesses            970
MEREP    Mère Paillard                 966
BOTTM    Bottom-Dollar Markets          956
KOENE    Königlich Essen              903
HANAR    Hanari Carnes                 839
LILAS    LILA-Supermercado            836
SEVES    Seven Seas Imports            818
RICSU    Richter Supermarkt            810
LEHMS    Lehmanns Marktstand           794
VAFFE    Vaffeljernet                  792
WARTH    Wartian Herkku                737
BLONP    Blondesdds1 père et fils      666
RICAR    Ricardo Adocicados            660
AROUT    Around the Horn               650
OTTIK    Ottilies Käseladen            639
PICCO    Piccolo und mehr              624
OLDWO    Old World Delicatessen        603
EASTC    Eastern Connection            569
Time taken: 0.36 seconds, Fetched: 31 row(s)
```

6. Generar un archivo .py que permita mediante Spark insertar en hive en la tabla products_sent, los datos del punto 3 y 4, de manera tal que se vean las columnas order_id, shipped_date, company_name, phone, unit_price_discount (unit_price with discount), quantity, total_price (unit_price_discount * quantity). Solo de aquellos pedidos que hayan tenido descuento.

Creamos en la base de datos **northwind_analytics**, la tabla **products_sent** en Hive

```
hive> DROP TABLE IF EXISTS northwind_analytics.products_sent;
OK
Time taken: 0.074 seconds
hive>
> CREATE EXTERNAL TABLE northwind_analytics.products_sent (
>   order_id          INT,
>   shipped_date       BIGINT,      -- ahora coincide con Parquet
>   company_name       STRING,
>   phone              STRING,
>   unit_price_discount DOUBLE,
>   quantity           INT,
>   total_price        DOUBLE
> )
> STORED AS PARQUET
> LOCATION '/tables/external/northwind_analytics/products_sent';
OK
Time taken: 0.19 seconds
hive>
```

Allí insertaremos los datos de la vista generada en pyspark.

Creamos el archivo **transform_load_5.py** en **home/hadoop/scripts**

```

hadoop@615cf53bef6c:~/scripts$ cat > transform_load_5.py
#####
# Archivo: transform_load_5.py
#####

# Importamos librerias

from pyspark.sql import SparkSession

def main():
    # 1) Crear la sesión de Spark con soporte Hive

    spark = (
        SparkSession
        .builder
        .appName("transform_load_5")
        .enableHiveSupport()
        .getOrCreate()
    )

    # 2) Rutas de entrada (parquet desde HDFS)
    df1 = spark.read.parquet("/sqoop/ingest/envios")
    df2 = spark.read.parquet("/sqoop/ingest/order_details")

    # 3) Vistas temporales

    df1.createOrReplaceTempView("lista_ordenes_empresa_v")
    df2.createOrReplaceTempView("detalles_ordenes_v")

    # 4) Transformaciones

    query = """
    SELECT
        e.order_id,
        e.shipped_date,
        e.company_name,
        e.phone,
        ROUND(d.unit_price * (1 - CASE
            WHEN d.discount <= 1 THEN d.discount
            ELSE d.discount / 100.0
        END), 2) AS unit_price_discount,

        d.quantity,
        ROUND(
            (d.unit_price * (1 - CASE
                WHEN d.discount <= 1 THEN d.discount
                ELSE d.discount / 100.0
            END)) * d.quantity
            , 2) AS total_price
    FROM lista_ordenes_empresa_v e
    JOIN detalles_ordenes_v d
    ON e.order_id = d.order_id
    WHERE (CASE
        WHEN d.discount <= 1 THEN d.discount
        ELSE d.discount / 100.0
    END) > 0
    """

    df3_load = spark.sql(query)

    # 5) Loads

    df3_load.write.mode("overwrite").parquet("/tables/external/northwind_analytics/products_sent")

    spark.stop()

if __name__ == "__main__":
    main()

hadoop@615cf53bef6c:~/scripts$

```

Cambiamos permisos:

```

hadoop@615cf53bef6c:~/scripts$ chmod 777 transform_load_5.py
hadoop@615cf53bef6c:~/scripts$ ls -l
total 96
drwxrwxr-x 3 hadoop hadoop 4096 Sep 29 19:46 Notebook
-rw-rw-r-- 1 hadoop hadoop 14298 Nov 1 15:16 QueryResult.java
-rw-rw-r-- 1 hadoop hadoop 670 Feb 28 2022 derby.log
-rwxrwxr-x 1 hadoop hadoop 272 Sep 22 01:14 ingest.sh
-r-xr-xr-x 1 hadoop hadoop 398 Sep 17 16:57 landing.sh
-rwxrwxrwx 1 hadoop hadoop 716 Oct 18 16:24 landing_2.sh
-rwxrwxrwx 1 hadoop hadoop 914 Oct 24 19:25 landing_3.sh
-rwxrwxrwx 1 hadoop hadoop 273 Sep 28 12:14 pyspark_jupyter.sh
-rwxrwxrwx 1 hadoop hadoop 668 Nov 1 15:12 query_db_2_hdfs_parquet.sh
-rwxrwxrwx 1 hadoop hadoop 549 Nov 1 16:00 query_db_2_hdfs_parquet_2.sh
-rwxrwxrwx 1 hadoop hadoop 486 Nov 1 16:31 query_db_2_hdfs_parquet_3.sh
drwxr-xr-x 2 hadoop hadoop 4096 Feb 9 2022 spark-warehouse
-rwxrwxrwx 1 hadoop hadoop 1089 May 9 2022 start-services.sh
-rwxrwxrwx 1 hadoop hadoop 1964 Oct 19 00:37 transform_load.py
-rwxrwxrwx 1 hadoop hadoop 2897 Oct 19 12:48 transform_load_2.py
-rwxrwxrwx 1 hadoop hadoop 3909 Oct 25 04:35 transform_load_3.py
-rwxrwxrwx 1 hadoop hadoop 1133 Nov 2 01:09 transform_load_4.py
-rwxrwxrwx 1 hadoop hadoop 1962 Nov 2 19:18 transform_load_5.py
-rwxrwxrwx 1 hadoop hadoop 1058 May 9 2022 transformation.py
-rwxrwxrwx 1 hadoop hadoop 2068 Oct 31 22:14 weather_str_transform.py
-rwxrwxrwx 1 hadoop hadoop 1747 Nov 1 01:13 yahoo_weather2kafka.py
hadoop@615cf53bef6c:~/scripts$

```

Luego chequeamos que el archivo funciona correctamente desde **pyspark** (antes de integrarlo al piplane)

```
hadoop@615cf53bef6c:~/scripts$ /home/hadoop/spark/bin/spark-submit --master local[*] \
--conf spark.> --conf spark.sql.warehouse.dir=/user/hive/warehouse \
> --conf spark.sql.catalogImplementation=hive \
> --files /home/hadoop/hive/conf/hive-site.xml \
> /home/hadoop/scripts/transform_load_5.py
```

El archivo funciona bien.

Verificamos que realmente se cargaron los datos en la tabla de Hive

```
hive> select * from products_sent;
OK
10643  873169200000  Alfreds Futterkiste  030-0074321  9.0  2  18.0
10643  873169200000  Alfreds Futterkiste  030-0074321  34.2  15  513.0
10643  873169200000  Alfreds Futterkiste  030-0074321  13.5  21  283.5
10835  885351600000  Alfreds Futterkiste  030-0074321  10.4  2  20.8
10952  890708400000  Alfreds Futterkiste  030-0074321  23.75  16  380.0
11011  892436400000  Alfreds Futterkiste  030-0074321  12.59  40  503.5
10507  861678000000  Antonio Moreno Taquería (5) 555-3932  10.84  15  162.56
10507  861678000000  Antonio Moreno Taquería (5) 555-3932  39.1  15  586.5
10535  864183600000  Antonio Moreno Taquería (5) 555-3932  17.55  5  87.75
10535  864183600000  Antonio Moreno Taquería (5) 555-3932  16.56  10  165.6
```

7. Realizar un proceso automático en Airflow que orqueste los pipelines creados en los puntos anteriores. Crear un grupo para la etapa de ingest y otro para la etapa de process. Correrlo y mostrar una captura de pantalla (del DAG y del resultado en la base de datos)

Creamos el DAG

```
hadoop@615cf53bef6c:~/airflow/dags$ cat > DAG_ejercicio_7.clase_9.py
#####
#
# DAG creado para el ejercicio 7 de la clase 9
# /home/hadoop/airflow/dags/DAG_ejercicio_7.clase_9.py
#
#####

# Importamos librerías
from datetime import timedelta
from airflow import DAG
from airflow.operators.bash import BashOperator
from airflow.operators.dummy import DummyOperator
from airflow.utils.dates import days_ago
from airflow.utils.task_group import TaskGroup
import os # <-- agregado para construir PATH

# Definimos los argumentos por defecto del DAG
args = {
    "owner": "airflow",
}

# Rutas extra donde suelen estar Sqoop/Hadoop/Hive dentro del contenedor
EXTRA_PATH = "/usr/lib/sqoop/bin:/home/hadoop/hadoop/bin:/home/hadoop/hive/bin"

with DAG(
    dag_id="ingest-transform-4",
    default_args=args,
    schedule_interval="@ 0 * * *", # diario a medianoche
    start_date=days_ago(1),
    catchup=False,
    dagrun_timeout=timedelta(minutes=60),
    tags=["ingest", "transform", "sqoop"],
) as dag:

    inicio_proceso = DummyOperator(task_id="inicio_proceso")

    finaliza_proceso = DummyOperator(task_id="finaliza_proceso")

    # Tareas Ingest

    with TaskGroup("Ingest", tooltip="Tareas de Extracción de Datos") as ingest_group:
        export_table_1 = BashOperator(
            task_id="export_table_1",
            # Usamos bash -lc para leer perfiles de login y heredamos PATH correcto por env
            bash_command="/usr/bin/env bash -lc 'set -e; pipefail; /home/hadoop/scripts/query_db_2_hdfs_parquet.sh'",
            env={
                "PATH": f"{EXTRA_PATH}:{os.environ.get('PATH', '')}",
                "HADOOP_HOME": "/home/hadoop/hadoop",
                "HIVE_HOME": "/home/hadoop/hive",
                "SQOOP_HOME": "/usr/lib/sqoop",
            },
        ),
        export_table_2 = BashOperator(
            task_id="export_table_2",
            bash_command="/usr/bin/env bash -lc 'set -e; pipefail; /home/hadoop/scripts/query_db_2_hdfs_parquet_2.sh'",
            env={
                "PATH": f"{EXTRA_PATH}:{os.environ.get('PATH', '')}",
                "HADOOP_HOME": "/home/hadoop/hadoop",
                "HIVE_HOME": "/home/hadoop/hive",
                "SQOOP_HOME": "/usr/lib/sqoop",
            },
        ),
        export_table_3 = BashOperator(
            task_id="export_table_3",
            bash_command="/usr/bin/env bash -lc 'set -e; pipefail; /home/hadoop/scripts/query_db_2_hdfs_parquet_3.sh'",
            env={
                "PATH": f"{EXTRA_PATH}:{os.environ.get('PATH', '')}",
                "HADOOP_HOME": "/home/hadoop/hadoop",
                "HIVE_HOME": "/home/hadoop/hive",
                "SQOOP_HOME": "/usr/lib/sqoop",
            },
        ),
    )

    inicio_proceso --> export_table_1
    export_table_1 --> export_table_2
    export_table_2 --> export_table_3
    export_table_3 --> finaliza_proceso
```

```
# Tareas Transform - Load (sin cambios)

with TaskGroup("Transformacion_Load", tooltip="Tareas de Transformación de Datos") as process_group:
    products_sent = BashOperator(
        task_id="processing_clientes_productos",
        bash_command="""
            set -e
            /home/hadoop/spark/bin/spark-submit \
            --master local[*] \
            --deploy-mode client \
            --files /home/hadoop/hive/conf/hive-site.xml \
            /home/hadoop/scripts/transform_load_5.py
        """,
    )

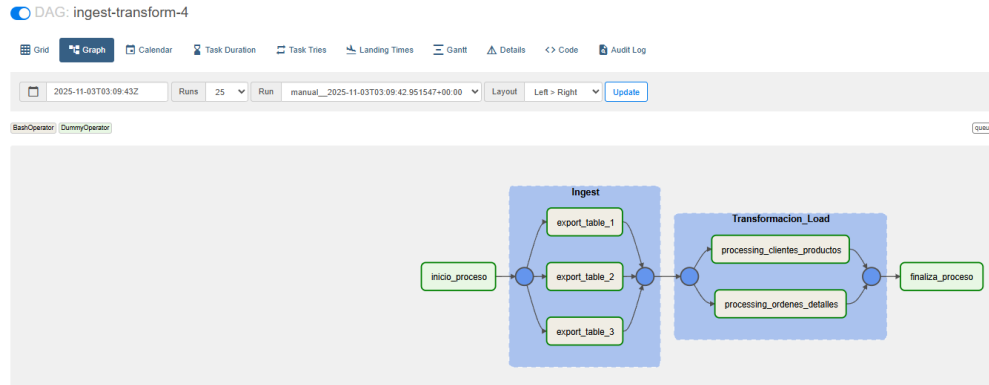
    products_sent = BashOperator(
        task_id="processing_ordenes_detalle",
        bash_command="""
            set -e
            /home/hadoop/spark/bin/spark-submit \
            --master local[*] \
            --deploy-mode client \
            --files /home/hadoop/hive/conf/hive-site.xml \
            /home/hadoop/scripts/transform_load_5.py
        """,
    )

# Inicio del flujo de tareas e ingestion
inicio_proceso --> ingest_group

# Ingestion y procesamiento en paralelo
ingest_group --> process_group

# Finalización del proceso de tareas
process_group --> finaliza_proceso
```

Grafica de Airflow:



Observamos que los datos están en las tablas del Warehouse:

DBeaver 24.1.4 - products_sold

Archivo Editar Navegar Buscar Editor SQL Base de Datos Ventana Ayuda

SQL Commit Rollback Auto localhost 2 default

Navegador d... northwind northwind <northwind> Repaso_SQL_queries.sql public

Propiedades Datos Diagrama ER

products_sold Enter a SQL expression to filter results (use Ctrl+Space)

	customer_id	company_name	productos_vendidos
1	SAVIA	Save-a-lot Markets	4.958
2	ERNSH	Ernst Handel	4.543
3	QUICK	QUICK-Stop	3.961
4	HUNGO	Hungry Owl All-Night Grocers	1.684
5	FRANK	Frankenversand	1.525
6	RATTC	Rattlesnake Canyon Grocery	1.383
7	FOLKO	Folk och få HB	1.234
8	HILAA	HILARION-Abastos	1.096
9	SUPRD	Suprêmes délices	1.072
10	WHITC	White Clover Markets	1.063
11	QUEEN	Queen Cozinha	1.031
12	BERGS	Berglunds snabbköp	1.001
13	BONAP	Bon app'	980
14	LINOD	LINO-Delicateses	970
15	MEREP	Mère Paillard	966
16	BOTTM	Bottom-Dollar Markets	956

DBeaver 24.1.4 - products_sent

Archivo Editar Navegar Buscar Editor SQL Base de Datos Ventana Ayuda

SQL Commit Rollback Auto localhost 2 default

Navegador d... northwind northwind <northwind> Repaso_SQL_queries.sql public customers orders <northwind> Script-26 orde

Propiedades Datos Diagrama ER

products_sent Enter a SQL expression to filter results (use Ctrl+Space)

	order_id	shipped_date	company_name	phone	unit_price_discount	quantity	total_price
1	10.643	873.169.200.000	Alfreds Futterkiste	030-0074321	9	2	18
2	10.643	873.169.200.000	Alfreds Futterkiste	030-0074321	34,2	15	513
3	10.643	873.169.200.000	Alfreds Futterkiste	030-0074321	13,5	21	283,5
4	10.835	885.351.600.000	Alfreds Futterkiste	030-0074321	10,4	2	20,8
5	10.952	890.708.400.000	Alfreds Futterkiste	030-0074321	23,75	16	380
6	11.011	892.436.400.000	Alfreds Futterkiste	030-0074321	12,59	40	503,5
7	10.507	861.678.000.000	Antonio Moreno Taqueria	(5) 555-3932	10,84	15	162,56
8	10.507	861.678.000.000	Antonio Moreno Taqueria	(5) 555-3932	39,1	15	586,5
9	10.535	864.183.600.000	Antonio Moreno Taqueria	(5) 555-3932	17,55	5	87,75
10	10.535	864.183.600.000	Antonio Moreno Taqueria	(5) 555-3932	16,56	10	165,6
11	10.535	864.183.600.000	Antonio Moreno Taqueria	(5) 555-3932	49,5	15	742,5
12	10.535	864.183.600.000	Antonio Moreno Taqueria	(5) 555-3932	18,9	50	945
13	10.677	875.242.800.000	Antonio Moreno Taqueria	(5) 555-3932	2,12	8	17
14	10.677	875.242.800.000	Antonio Moreno Taqueria	(5) 555-3932	26,55	30	796,36
15	10.453	856.926.000.000	Around the Horn	(171) 555-7788	9,18	15	137,7
16	10.453	856.926.000.000	Around the Horn	(171) 555-7788	10,8	25	270
17	10.707	877.575.600.000	Around the Horn	(171) 555-7788	12,75	28	357
18	10.741	879.822.000.000	Around the Horn	(171) 555-7788	15,2	15	228
19	10.743	880.081.200.000	Around the Horn	(171) 555-7788	11,4	28	319,2
20	10.953	890.794.800.000	Around the Horn	(171) 555-7788	11,87	50	593,75
21	10.953	890.794.800.000	Around the Horn	(171) 555-7788	76,95	50	3.847,5
22	10.572	867.207.600.000	Berglunds snabbköp	0921-12 34 65	28,8	10	288

